MSc. in Computing
Practicum Approval Form

Student Details

Project Title:	Automated Multi-Modal Privacy Protection Framework for Personal Image Data Sharing
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Date of Submission	November 21st, 2025

About the Practicum

What is the topic of your proposed practicum?

This practicum develops an automated privacy protection framework for personal image data, focusing on face and text de-identification with optional screen extension if time permits. We will implement and compare multiple detection and anonymization techniques including StyleGAN2, DeepPrivacy2, and diffusion-based approaches. The core innovation is an adaptive method selection mechanism that automatically chooses optimal anonymization based on image characteristics, optimizing privacy-utility trade-offs for egocentric lifelog imagery. By systematically comparing methods on real-world wearable camera data, this work addresses identified gaps in current privacy protection research. The project aims to deliver proof-of-concept benchmarks and validation software, providing practical insights for privacy-preserving data sharing.

Please provide details of the papers you have read on this topic:

1. Khorzooghi, S. M. S. M., & Nilizadeh, S. (2023). Examining StyleGAN as a Utility-Preserving Face De-identification Method. Proceedings on Privacy Enhancing Technologies, 2023(4), 587-605. https://doi.org/10.56553/popets-2023-0114

2. Meden, B., Mallick, S., Corneanu, C. A., Peer, P., Escalera, S., & Emeršič, Ž. (2023). Face deidentification with controllable privacy protection. Image and Vision Computing, 133, 104678. https://doi.org/10.1016/j.imavis.2023.104678

3. Hukkelås, H., Lindseth, F., & Mester, R. (2023). DeepPrivacy2: Towards Realistic Full-Body Anonymization. Proceedings of the IEEE/CVF Winter Conference on Applications of Computer Vision (WACV), 1329-1338. https://doi.org/10.1109/WACV56688.2023.00138

4. Khan, W., Topham, L. K., Breckon, T. P., & Tam, P. C. H. (2025). Person De-Identification: A Comprehensive Review of Methods, Datasets, Applications, and Ethical Aspects Along with New Dimensions. IEEE Transactions on Biometrics, Behaviour, and Identity Science, 7(3), 293-312. https://doi.org/10.1109/TBIOM.2024.3485990

5. He, X., Guo, W., Tan, Z., Yang, X., Xie, J., Yu, D., & Cao, X. (2024). Diff-Privacy: Diffusion-Based Face Privacy Protection. IEEE Transactions on Circuits and Systems for Video Technology, 34(12), 12677-12690. https://doi.org/10.1109/TCSVT.2024.3449290

6. Li, Y., Zhang, G., Cheng, J., Li, Y., Shan, X., Gao, D., Lyu, J., Yuan, L., Bi, N., & Vasconcelos, N. (2025). EgoPrivacy: What Your First-Person Camera Says About You? Proceedings of the 42nd International Conference on Machine Learning (ICML), PMLR 267, 36794-36809. https://doi.org/10.48550/arXiv.2506.12258

How does your proposal relate to existing work on this topic described in these papers?

Our proposal builds upon foundational de-identification research while addressing specific gaps in face anonymization for egocentric data. StyleGAN-based methods (Khorzooghi & Nilizadeh, 2023) establish privacy-utility trade-offs through style mixing at different network layers. Controllable privacy approaches (Meden et al., 2023) demonstrate feasibility of adaptive anonymization strength, which we extend through automated context-based selection. DeepPrivacy2 (Hukkelås et al., 2023) achieves high-quality anonymization without requiring facial key point detection. Diffusion-based approaches (He et al., 2024) represent latest generative techniques achieving natural-looking privacy protection.

Khan et al.'s comprehensive review (2025) confirms that while face de-identification has advanced significantly, systematic comparison and adaptive method selection for egocentric imagery remain unexplored. The 2025 EgoPrivacy benchmark (Li et al.) reveals severe privacy risks in first-person camera data, with foundation models achieving 70-80% identification accuracy even in zero-shot settings. However, current methods are typically evaluated on controlled datasets rather than challenging egocentric scenarios.

Our contribution systematically implements and compares multiple face anonymization methods on egocentric data, validating adaptive method selection. Rather than proposing entirely new algorithms, we integrate existing techniques into a unified evaluation framework optimized for wearable camera imagery. We also incorporate text de-identification, with optional screen extension if time permits. This practical approach bridges current research and real-world deployment needs, providing empirical evidence for privacy-utility trade-offs in egocentric settings. The adaptive selection mechanism represents our novel contribution, building on Meden et al.'s controllable privacy concept.

What are the research questions that you will attempt to answer?

RQ1: Face Detection Performance

How do YOLOv8-Face and RetinaFace perform on egocentric lifelog imagery compared to controlled datasets? We will measure precision, recall, F1-score, and processing speed to establish a performance baseline for egocentric views.

RQ2: Face Anonymization Method Comparison
How do StyleGAN2, DeepPrivacy2, and diffusion-based approaches compare in privacy protection and utility preservation on egocentric data? Privacy will be measured via re-identification resistance (target: <20% identification rate). Utility measured via attribute accuracy (target: >75%).

RQ3: Adaptive Method Selection Effectiveness
Does context-aware method selection based on image quality and characteristics, achieve better privacy-utility balance than fixed-strategy anonymization? We will compare processing speed and privacy outcomes between adaptive and fixed approaches.

RQ4: Text and Optional Screen De-identification
How effectively can text de-identification be integrated with face anonymization? If time permits, we will extend to screen detection and redaction. This multi-modal approach addresses gaps identified by Khan et al. (2025), with screen processing positioned as optional extension based on timeline feasibility.

What software and programming environment will you use?

We will use Python 3.10+ with PyTorch 2.0+ as our deep learning framework. For face detection, we will implement YOLOv8-Face as the primary method with RetinaFace for accuracy comparison. Our anonymization pipeline integrates StyleGAN2 (Khorzooghi & Nilizadeh, 2023), DeepPrivacy2 (Hukkelås et al., 2023), and diffusion-based approaches (He et al., 2024). The adaptive selection mechanism automatically chooses between these methods based on image quality metrics.

Computational infrastructure leverages the local RTX 5060 GPU laptop (8GB VRAM) as primary resource for development and processing. Google Colab provides supplementary GPU capacity for parallel batch processing. Key Python libraries include OpenCV (image processing), torchvision (data transformations), EasyOCR (text detection), facenet-pytorch (face embeddings for privacy evaluation), scikit-learn (metrics), and Weights & Biases (experiment tracking). The modular architecture enables proof-of-concept development with potential for future extension. All code will be version-controlled on GitLab with documentation for reproducibility.

What coding/development will you do?

We will develop a modular pipeline with three main components:

1. **Detection Module:** Implements YOLOv8-Face for face localization with RetinaFace comparison. Preprocessing handles image normalization and optional deblurring for challenging egocentric images. Text detection via EasyOCR. Optional screen detection using YOLOv8 if timeline permits.
2. **Anonymization Module:** Integrates StyleGAN2, DeepPrivacy2, and diffusion-based methods. Each method processes detected faces and replaces them while preserving background context. Text redaction applies blurring or inpainting to identified text regions. Baseline blur and pixelation techniques provided for reference.
3. **Evaluation and Adaptive Selection Module:** Analyzes image characteristics (lighting quality, blur level, viewing angle) to automatically select the most appropriate anonymization method. This implements our core novel contribution, adaptive method selection based on image quality rather than applying fixed strategy to all images.

All code will be version-controlled on GitLab with clear documentation and modular structure enabling future researchers to integrate additional detection or anonymization methods.

What data will be used for your investigations?

Our investigations utilize the DCU NTCIR lifelog collection comprising 90,000+ egocentric wearable camera images with associated metadata (timestamps, visual concept annotations). This dataset captures diverse daily activities with challenging characteristics: variable lighting, motion blur, and extreme viewing angles typical of first-person perspective.

For baseline comparison, we use high-quality face datasets: CelebA-HQ (30,000 images) and FFHQ (70,000 images). Testing methods on controlled data first establishes baselines before evaluating on difficult egocentric imagery.

Privacy evaluation employs standard face recognition models (facenet-pytorch, ArcFace) to measure re-identification risk. Utility evaluation uses attribute classifiers (age, gender, expression) and image quality metrics (PSNR, SSIM, FID). This multi-faceted approach enables rigorous comparison of privacy-utility trade-offs.

Text regions will be identified in a subset of lifelog images for text de-identification validation. Optional screen annotation (500-1000 images) may be pursued if time permits for extended multi-modal evaluation.

Is this data currently available, it not, where will it come from?

The NTCIR lifelog datasets are provided through DCU Insight Centre, where Dr. Zhou co-organizes the NTCIR lifelog tasks, ensuring institutional access and eliminating data acquisition risks. CelebA-HQ and FFHQ are publicly available datasets. All necessary data is currently accessible.

What experiments do you expect to run?

We expect to run four major experimental phases:

1. **Face Detection Baseline:** Compare YOLOv8-Face and RetinaFace on egocentric lifelog images measuring precision, recall, F1-score, and processing speed. Establish detection accuracy baseline on challenging first-person perspective data.
2. **Anonymization Method Comparison:** Evaluate StyleGAN2, DeepPrivacy2, and diffusion approaches on 500 FFHQ images (easy baseline) and 500 NTCIR lifelog images (challenging real-world). Measure privacy (re-identification rates via facenet/ArcFace), utility (attribute accuracy), and processing speed. Adjusted targets: <20% re-identification rate (acknowledging egocentric difficulty), >75% attribute accuracy (realistic for challenging conditions).
3. **Adaptive Method Selection Validation:** Compare fixed anonymization strategy (always use best method) versus adaptive selection (automatically choose method per image). Metrics: processing speed, privacy-utility balance, memory efficiency.
4. **Text De-identification and Optional Screen Extension:** Evaluate text detection and redaction effectiveness on lifelog subset. If time permits, implement and evaluate screen detection and anonymization. Multi-modal integration demonstrates comprehensive privacy protection addressing gaps identified by Khan et al. (2025).

What output do you expect to gather?

We expect to gather: (1) quantitative privacy metrics including face re-identification resistance rates (target <20%) and embedding distance measurements; (2) utility preservation scores via attribute classification accuracy (target >75% for age, gender, expression); (3) image quality metrics (PSNR, SSIM, FID); (4) computational efficiency benchmarks (FPS, memory usage, total processing time); (5) comparative analysis tables and visualizations ranking anonymization methods across privacy-utility trade-offs; (6) validation of adaptive method selection effectiveness; (7) text de-identification performance metrics; and (8) proof-of-concept implementation with documentation. These outputs enable systematic comparison of face anonymization approaches on egocentric data, demonstrate practical feasibility, and identify remaining challenges for future extensions.

How will the results be evaluated?

Results will be evaluated through a multi-dimensional framework: (1) privacy protection measured via face recognition attack success rates using facenet-pytorch and ArcFace (target <20%); (2) utility preservation via attribute classification accuracy for age, gender, and expression (target >75); (3) image quality metrics (PSNR, SSIM, FID scores); (4) computational efficiency (FPS, memory consumption, total processing time); (5) comparison of adaptive versus fixed method selection strategies; (6) text de-identification effectiveness; and (7) qualitative analysis of challenging egocentric scenarios. These metrics provide comprehensive assessment of privacy-utility trade-offs and practical deployment feasibility.